Preliminary statements and their precise scope

Human Compression project

Abstract

We record the complete preliminary dependency boundary for the fixed 1999 rank-three Suzuki article. Several literal arithmetic assertions need correction: integer divisibility is insufficient for subfield divisibility, the residue class controlling the factor divisible by five is misstated, and a prime extension must remain in the same torus. We give small exact witnesses and the corrected general statements. The standard group structure and the explicitly cited linear-diagram theorem remain attributed inputs.

1 Definitions and quoted structural facts

Throughout, the source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63, in the fixed twenty-one-page edition. Write

$$q = 2^{2e+1}, \qquad \theta = 2^{e+1}, \qquad \alpha_q = q + \theta + 1, \quad \beta_q = q - \theta + 1.$$

A simple Suzuki subfield group has size parameter at least eight. The source uses $a \mid_M b$ to mean that b/a is prime, and $a \mid_P b$ to mean that a is a proper divisor of b (printed p. 47). These symbols must not be replaced by ordinary divisibility in a source join.

Source Lemma 2.1 states that a rank-three coset pregeometry is a geometry. Indeed an incident pair of cosets has a common representative g, and adding gG_k of the missing type gives a chamber. A singleton similarly extends using any representative; the empty flag extends to the base chamber. Thus every flag extends to a chamber. No flag-transitivity assertion follows from this argument.

The classical Suzuki maximal-subgroup list in source Table 1 is an explicitly attributed premise from Lüneburg and Suzuki. Source Lemmas 3.1 and 3.2 are explicitly quoted from the earlier rank-two paper. Their required content is independently derived in the accepted L115 preliminary note: the maximal subgroups of $B_q, D_q, N_+(q), N_-(q)$, and pairwise coprimality of $q-1, \alpha_q, \beta_q$. The resulting bounds on pairwise intersection orders, respectively $2, 2, 4$, follow from divisibility of subgroup orders. No additional finite census is used here.

Source Lemma 3.8 is expressly stated to follow from the main theorem of M. Aschbacher, *Flag structures on Tits geometries*, Geometriae Dedicata 14 (1983), 21–31. We preserve this as a cited premise in its source form: a rank-three geometry with a linear diagram is flag transitive. This note does not reconstruct the cited theorem. In particular, no arbitrary intersection pattern is declared to have such a diagram without the relevant residue and diagram hypotheses.

2 Field degrees and the arithmetic statements

The classical list gives one conjugacy class of maximal simple subfield groups for each prime $p \mid n$, where $q = 2^n$, provided $n/p \geq 3$. If n is prime there are none. If n is composite all these quotients are at least three. Consequently there is exactly one class if and only if $n = p^k$, $k \geq 2$. This proves source Lemma 3.3.

The repunit argument in the accompanying normalizer-restriction note proves the implication in Lemma 3.4: if $\mathrm{Sz}(s)$ is maximal in $\mathrm{Sz}(q)$ and $(q-1)/(s-1)$ is prime, then $q = s^p$, $s = 2^{p^a}$ for a prime p and a positive integer a. Therefore, if there are at least two maximal subfield classes, $s-1$ is *not a maximal proper divisor* of $q-1$. This is the consequence needed in the source.

Lemma 3.5: the printed word "prime". Printed p. 48 instead says that $s-1$ is not a prime divisor of $q-1$. That literal assertion is false. Take $q = 2^{15} = 32768$ and $s = 8$. The subfield group $\mathrm{Sz}(8)$ is maximal, of field index five, and there is also the different maximal class $\mathrm{Sz}(32)$, of field index three. Yet $s-1 = 7$ is prime and

$$32767 = 7 \cdot 31 \cdot 151.$$

Its quotient $32767/7 = 4681$ is composite, in agreement with the corrected maximal-divisor assertion. This is a wording correction to an ancillary lemma, not an additional principal-refutation count.

Lemma 3.6: divisibility and the residue variable. Part (1) is $5 \mid q^2+1$, not $s^2+1 \mid q^2+1$. It follows at once from $q^2 = 2^{4e+2} \equiv -1 \pmod 5$. For part (2), the native statement assumes merely $s \geq 8$ divides q. This is insufficient. At $s = 8, q = 32$ it holds as integer divisibility, but

$$(\alpha_8, \beta_8) = (13, 5), \qquad (\alpha_{32}, \beta_{32}) = (41, 25),$$

and 13 divides neither 41 nor 25. The corrected hypothesis is $s = 2^d$, $d \mid 2e+1$, equivalently $q = s^r$ for an odd integer r. Under that hypothesis the compatible-torus divisibilities, with the possible interchange of the two signs, are proved in the accepted L115 preliminary note. Only that version is imported in our proofs.

Part (3) prints the residue class of $2e+1$ modulo four. The correct variable is e:

$$5 \mid \alpha_q \iff e \equiv 0, 3 \pmod 4, \qquad 5 \mid \beta_q \iff e \equiv 1, 2 \pmod 4.$$

To prove this for all e, observe that both 2^{2e+1} and 2^{e+1} are periodic modulo five with period four in e. For $e = 0, 1, 2, 3$, the respective residues of (α_q, β_q) are $(0,1), (3,0), (1,0), (0,3)$. The printed formula already fails at $q = 8$, where $2e+1 = 3$ but $\alpha_q = 13$ is not divisible by five.

Lemma 1. *For odd positive integers m, n,* $\gcd(2^{2m}+1, 2^{2n}+1) = 2^{2\gcd(m,n)}+1$.

Proof. Put $x = 4$ and $d = \gcd(m, n)$. The Euclidean algorithm for powers gives $\gcd(x^{2m}-1, x^{2n}-1) = x^{2d}-1$. The common divisor $D = \gcd(x^m+1, x^n+1)$ therefore divides $(x^d-1)(x^d+1)$. It is coprime to x^d-1: the latter divides x^m-1, and an odd common divisor of x^m-1, x^m+1 is one. Hence $D \mid x^d+1$. Conversely $m/d, n/d$ are odd, so x^d+1 divides both x^m+1 and x^n+1. This proves the equality and source Lemma 3.7. □

3 Unique extensions and proper overgroups

For source Lemma 3.9(i), let $H = C_d \rtimes C_2$, $d > 1$ dividing q^2+1. The cyclic odd subgroup determines one ambient nonsplit torus T and its normalizer $N = T \rtimes C_4$. Every group of the form $C_d \rtimes C_4$ containing H normalizes its characteristic odd subgroup and hence lies in N. An involution of N is contained in exactly one cyclic subgroup of order four: that cyclic subgroup is its centralizer in N, since the involution inverts T. Adjoining this C_4 to the unique order-d subgroup of T gives the unique extension. This proves (i), including actual existence.

For (ii), write $|T| = a_\epsilon(q)$. If $H = C_d \rtimes C_4$ and $p \mid |T|/d$ is prime, the unique subgroup of T of order dp, together with the given C_4, supplies a unique $C_{dp} \rtimes C_4$ containing H. Every possible extension must have this torus and this cyclic odd subgroup, proving uniqueness.

Lemma 3.9(ii): the required compatible factor. The native condition is instead $p \mid (q^2+1)/d$, which can choose a prime from the opposite, coprime nonsplit factor. At $q = 8, d = 5, p = 13$, the subgroup $C_5 \rtimes C_4$ exists, but $C_{65} \rtimes C_4$ does not: it would contain an element of order65, whereas the odd elements of Sz(8) lie in cyclic tori of orders $5, 7, 13$. The source condition therefore needs the compatible-factor restriction just proved. The retained torus partition is the only group-theoretic premise in this small obstruction; no enumeration is required.

The full split-torus overgroup argument in the normalizer-restriction note gives the precise version of source Lemma 3.10: the *strict proper* overgroups of C_{q-1} are its dihedral normalizer, the two point stabilizers, and their two $E_q \rtimes C_{q-1}$ subgroups. The torus itself and the whole ambient group must be added if neither strictness nor properness is imposed. We state this qualification explicitly rather than assigning an unintended meaning to "containing".

Finally let $S = \mathrm{Sz}(s)$ be maximal in $G = \mathrm{Sz}(q)$ and let $H = D_{2(s-1)}$ be its full split normalizer. Every proper overgroup of H is either a dihedral subgroup D_{2u} of its ambient split normalizer, with $s-1 \mid u \mid q-1$, or a simple subfield group $\mathrm{Sz}(t)$. Point stabilizers are excluded by the inversion argument, and nonsplit normalizers by coprimality. In the subfield case the rotation subgroup is split, so $s-1 \mid t-1$. Writing $s = 2^d, t = 2^f$ gives $d \mid f$; this follows from the Euclidean identity $\gcd(2^d-1, 2^f-1) = 2^{\gcd(d,f)}-1$. Maximality of the field degree d in the ambient degree forces $f = d$ for a proper subfield. The accepted uniqueness of the subfield containing this embedded full split normalizer then gives $\mathrm{Sz}(t) = S$. This proves source Lemma 3.11 for proper overgroups. The ambient group itself remains excluded by that explicit qualification.

4 Attribution and remaining work

The new source joins above distinguish literal errors, intended corrected statements, and earlier quoted inputs. They do not replace the primary classification counterexamples already accepted for L114, and add no principal-refutation or ordinary-recovery count. The arithmetic checks attached to this note only verify the displayed small integers; the general proofs and imported subgroup premises require distinct mathematical review. The source's general rank-three thinness assertion under its literal algebraic local-action convention and the final source-case joins remain separate obligations. No Lean verification or compression measurement is claimed.